LEAKLISTEN · MARKET AND VIABILITY

LeakListen market and viability

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DOCUMENT CONTROL

Document ID	LKL-MKT-001	Version	0.1
Status	Draft	Date	2026-11-01
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
Source	docs/10-market.md @ a81e84a		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First market and viability note (prices checked 2026-11-01); commercial benchmarks are few because noise logger prices are mostly quoted privately

LeakListen market and viability

LeakListen is for small and mid-sized water utilities, private networks and researchers who cannot afford a closed fleet of noise loggers. Published prices are scarce; the two found are far apart and neither is a like-for-like comparison, so viability is argued from the parts cost (USD 142 against a USD 130 target) rather than a market price. Prices were checked on 2026-11-01. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Small and mid-sized water utilities	Leakage teams in the UK and Europe, first candidate region the UK (LKL-DEC-001)	SubSurface Enigma 3 Hybrid Logger System: USD 22,500 current price (USD 27,000 list) for three magnet-base loggers, three cable loggers, three hydrophone kits, three stainless 2.5 inch hydrant caps, a USB cable and cases; about USD 3,750 per logger if the rest is free	142 in parts for option A (bom/bom.csv), no labour or board design	A fleet of open loggers at a fraction of a closed kit; data and radio open (UR-010, UR-015)
Community water schemes and private networks	Campuses, industrial parks, airports	An Indian listing for an "RD521 Noise Logging System" at Rs 1,900 per piece, excluding taxes, rated for 10 years of battery and a 5-year warranty (about USD 22 at Rs 85 per USD, an estimate not checked); what a piece includes is not stated	142	The low end is a single unit price of unknown scope and cannot be compared; LeakListen adds LoRaWAN reporting and open data
Researchers and students	Leak noise research, teaching	Closed commercial loggers; no open dataset found in this search	142	An open, documented logger and dataset (UR-015)

The one hard benchmark (Enigma 3) shows what a utility pays today for magnet-base loggers plus hydrophones on stainless hydrant caps, which is the pairing of options A and B in this repo. The price is for a kit with host software; whether software and a receiver are included was not stated on the page. The benchmark does not show how many such kits utilities buy, and no market size is claimed here.

Open route

A lab or a university builds the logger from the drawings, bill of materials and build plan; a utility or a research partner places it; the lab's TwinKit gateway or any LoRaWAN network server receives the 24-byte summary. CERN-OHL-S keeps changes open. The main board design (TRL 4) and a utility partner (LKL-RSK-001 P1) are the two gates to a pilot.

Sources

- SubSurface Leak Detection (listing on Tiger Supplies). "SubSurface Enigma 3 Hybrid Logger System." https://tigersupplies.com/products/subsurface-enigma-3-hybrid-logger-system (checked 2026-11-01): current price USD 22,500.00, list USD 27,000.00; contents as stated in the text.

- IndiaMART listing. "RD521 Noise Logging System." https://m.indiamart.com/proddetail/rd521-noise-logging-system-4246181233.html (checked 2026-11-01): Rs 1,900 per piece, excluding all taxes.
- Gutermann, https://en.gutermann-water.com/, cited in LKL-PRB-001 (a commercial correlating logger, no price found).
- The exchange rate of Rs 85 per USD is an estimate, not checked.